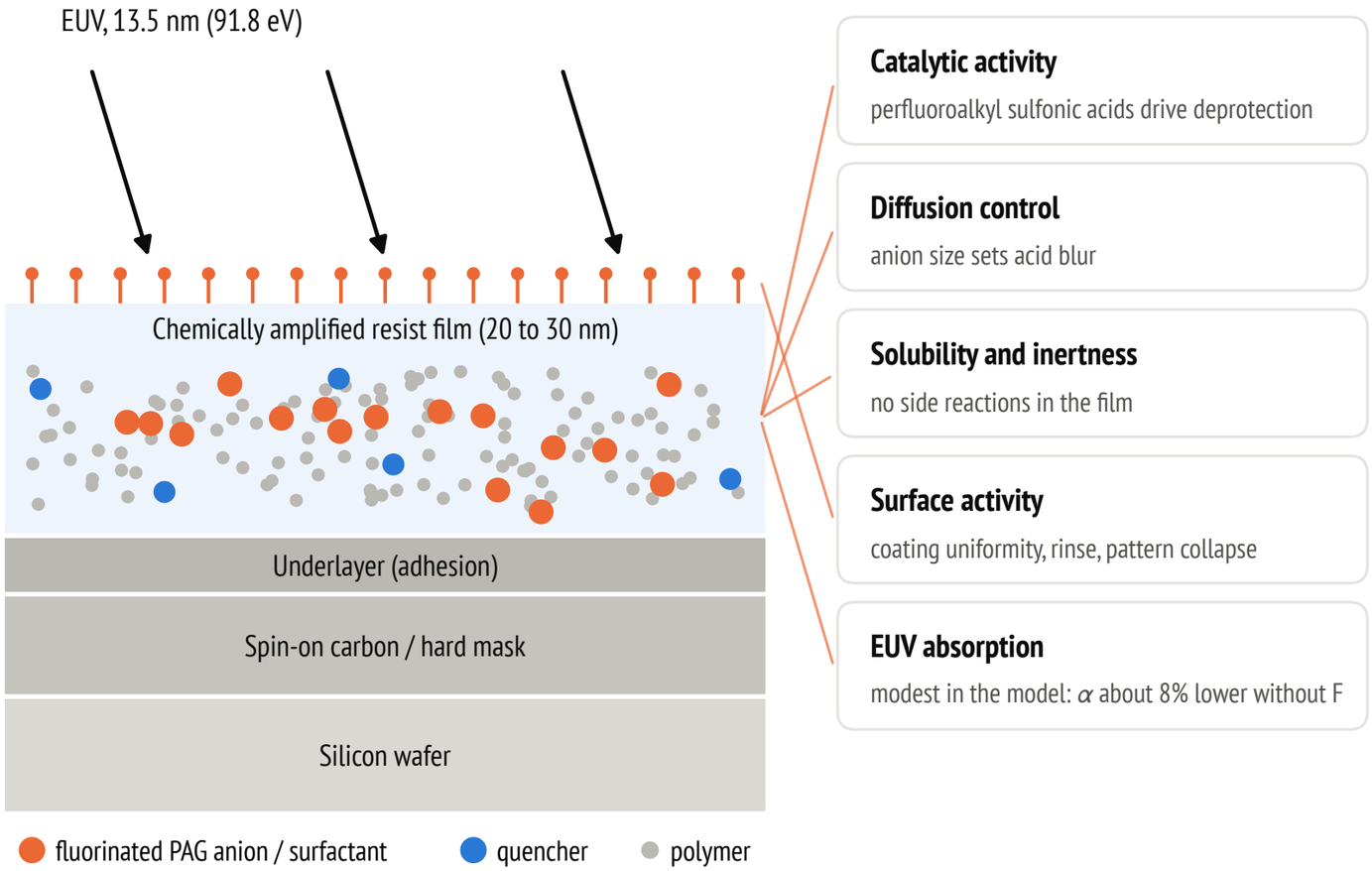
a



Catalytic activity
perfluoroalkyl sulfonic acids drive deprotection

Diffusion control
anion size sets acid blur

Solubility and inertness
no side reactions in the film

Surface activity
coating uniformity, rinse, pattern collapse

EUV absorption
modest in the model: α about 8% lower without F

b

Photon budget — Fig. 2
Henke cross sections, Eqs. 1 to 3

Stochastic patterning — Figs. 3, 4
verified solver; activity criterion, Eqs. 4 to 17

Shrinking-pitch stress test — Fig. 5
fixed aerial image; NA labels are proxies

Design-space exploration — Fig. 6
formulation families; fluorine as objective

Screening inventory — Fig. 7
allocation to waste pathways, not emission

c

